Milestone 3.2 Report: Thermal-Hydraulics Analysis of Core Load-Following Operation

A. J. H. Lee, T. Kozlowski, K. D. Huff

UIUC-ARTS-2020-02

July 31, 2020

ANALYSIS OF REACTOR TRANSIENTS AND STABILITY (ARTS)

DEPT. OF NUCLEAR, PLASMA, & RADIOLOGICAL ENGINEERING
UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

This research is being performed using funding from the Department of Energy ARPA-E MEITNER Program (award DE-AR0000983) and made use of the resources of the High Performance Computing Center at Idaho National Laboratory, which is supported by the Office of Nuclear Energy of the U.S. Department of Energy and the Nuclear Science User Facilities under Contract No. DE-AC07-05ID14517, as well as the Blue Waters sustained-petascale computing project, which is supported by the National Science Foundation (awards OCI-0725070 and ACI-1238993) and the state of Illinois. Blue Waters is a joint effort of the University of Illinois at Urbana-Champaign and its National Center for Supercomputing Applications.

1. Abstract

The University of Illinois, Urbana-Champaign (UIUC) performed a study in August 2019 to analyze the neutronics behavior of the Transatomic Power Molten Salt Reactor (TAP MSR) core during load-following operations, culminating in the Milestone 3.1 Report [1]. As a follow-up to the previous work, we will be analyzing the thermal-hydraulics behavior of the TAP MSR in this work.

This work is a deliverable to Milestone 3.2 of the DOE ARPA-E MEITNER project (DE-AR0000983) with the following objectives:

1) Perform analysis of core thermal-hydraulics to predict local flow, temperature and pressure at different power levels and control rod positions.
2) Determine whether core material and safety constraints can be met while taking sufficient measures to avoid exceeding such constraints.

2. Introduction

In the previous report, we had obtained the neutronics behavior of the TAP MSR by simulating the fuel salt flow as a uniform advection. As an extension to the previous work, due to the complex internal geometry of the reactor core, it is worthwhile to determine whether the variation of the local flow from uniform advection would affect the safety and load-following behavior of the reactor. Therefore, in this work, we had simulated the fuel salt flow using the Incompressible Navier-Stokes physics submodule in MOOSE with the heat generation profile determined using Moltres to determine the thermal-hydraulics behavior of the TAP MSR.

2.1 Core Geometry, Material Properties and Core Constraints

The geometry of the TAP MSR core used in this work is the same as that used in the Milestone 3.1 report. The moderator rods and control rods are arranged in regular lattice patterns as shown in Figure 1, with 347 moderator rods per quarter core. The RZ cross section of the reactor is shown in Figure 2. The properties of the materials present in the core, as reproduced from the previous report, are presented in Table 1.

Table 1: Tabulation of the material property values used in the simulations in this study.

	Material	Density ($kgcm^{-3}$)	Heat Capacity, c_p ($JK^{-1}kg^{-1}$)	Thermal Conductivity, k ($Wcm^{-1}K^{-1}$)
Vessel Wall	Hastelloy	8.86 E-03	395.8[1]	0.209 [2]
Fuel Salt[5]	LiF-(Act)-F_4	(6.105-0.001272T) E-03 [3]	746.66 [4]	0.05[2]
Moderator	$ZrH_{1.66}$	5.66 E-03	726.744 [5]	0.1975[3] [5]
Moderator Cladding	SiC	3.21 E-03	1200 [6]	0.77834 [6]
Control Rod	B_4C	2.52 E-03	1800 [7]	0.1[4] [7]
Air	Air	3.53 E-04 / T [8]	1120.9 [8]	6.27 E-04 [8]

[1] Estimated using Dulong-Petit Law.
[2] Thermal conductivity was estimated. T in the density formula refers to temperature in Kelvins.
[3] Thermal conductivity and heat capacity estimated at 900 Kelvins.
[4] Conservative estimates for thermal conductivity and heat capacity.
[5] The viscosity of the fuel salt used in this work was estimated from **[3]** to be 0.0209 Pa·s.

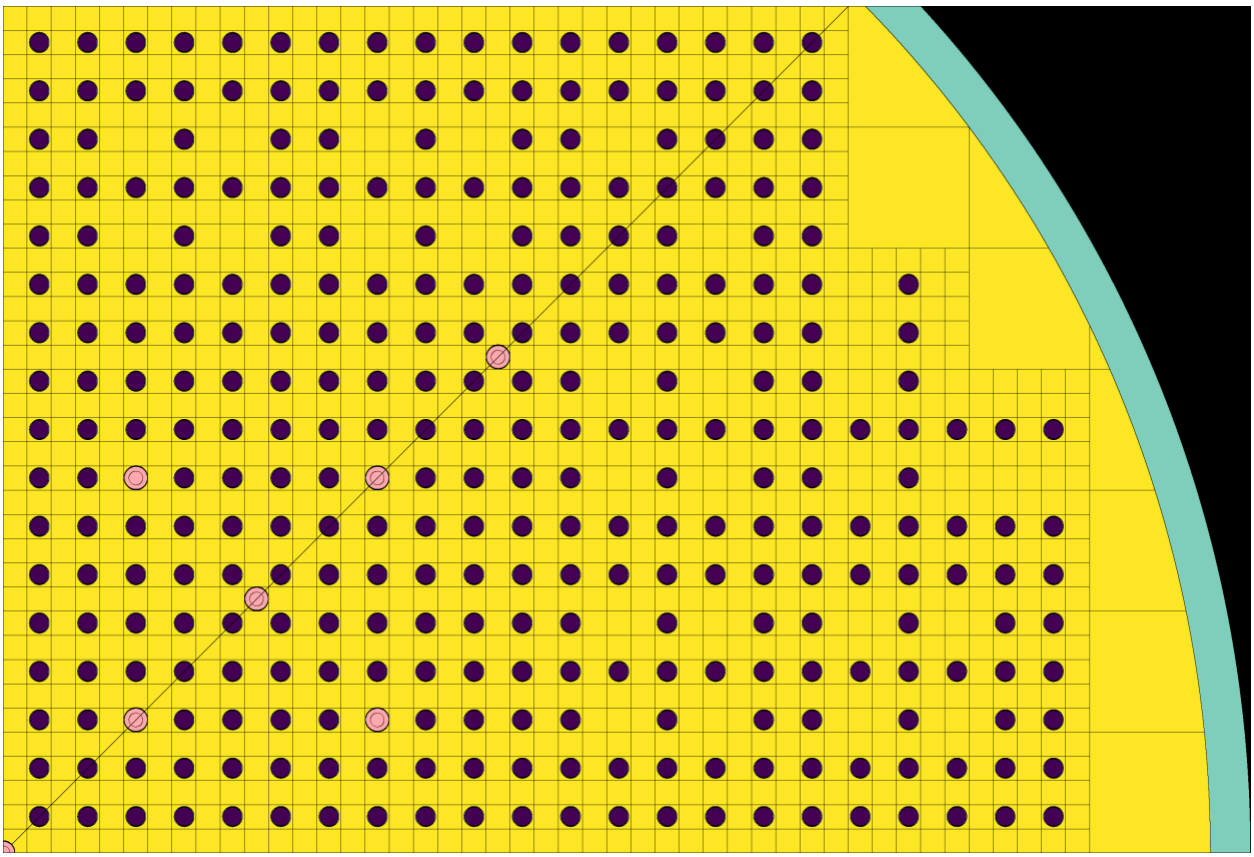

Figure 1: Axial cross section of the TAP MSR core illustrating the lattice like pattern of the moderator (violet), control rods (pink) and fuel salt (yellow).

The reactor dimensions used in this study are presented in Table 2 [9]. The central 300 cm portion of the reactor where the control rods can travel between the parked (withdrawn) position and fully inserted position is the active region of the reactor. Axially above and below the active region are two 25 cm plenum regions, making the total reactor height to be 350 cm [Figure 2].

Table 2: Dimensions of the reactor, referenced and extended from the Oak Ridge National Lab Report (ORNL/TM-2017/475) **[9]**.

Component	Parameter	Value (cm)	
Moderator Rod	Cladding thickness	0.10	
	Radius	1.15	
	Length	300	
	Pitch	3.0	
Control Rod	Radius	1.25	
	Length	325	
	0% Insertion Position (Parked)	25	from top edge
	100% Insertion Position	325	from top edge
Core	Inner Radius	150	
	Vessel Wall Thickness	5	
	Plenum Height	25	
	Active Reactor Region	300	
	Total Reactor Height	350	

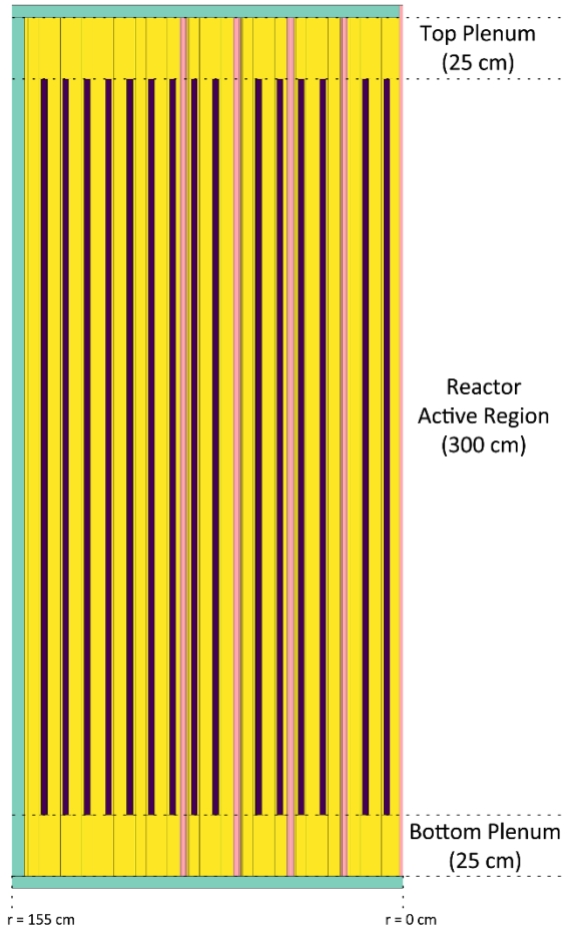

Figure 2: 45° RZ cross section of the TAP MSR core denoting the top and bottom plenums as well as the active reactor region. The image is a view of the TAP MSR from a 45 degree angle similar to a view from the diagonal line of Figure 1.

Similar to the previous report, we will be using a conservative material temperature constraint of 900 °C that corresponds to the temperature limit of the Hastelloy components since Hastelloy was identified as the material with the lowest temperature limit. Nonetheless, it should be noted that Hastelloy is used as a bulk structural material and it may be possible to substitute Hastelloy with other more resilient materials for certain localized regions with challenging environments.

The core inlet temperature was set at 550 °C, with a target outlet temperature of around 650 °C when the reactor was operating at the rated power level of 1250 MW_{th}. To avoid exceeding this target outlet temperature, we had set the salt flow rate at 65 cm/s, which was a slight increase from that used in the previous work (53 cm/s).

3. Thermal-Hydraulics Core Simulation

The thermal-hydraulics simulation of the TAP MSR core was performed using the following key steps:

1) We first performed a steady state simulation of the fuel salt flow using the Incompressible Navier-Stokes (INS) method in MOOSE. Because the control rods in the TAP MSR are housed in guide-tubes, the fuel salt flow is largely isolated from the varying control rods positions. This means that one single salt flow simulation can be reused for different control rods configurations, assuming that the differences in neutronics behavior do not significantly affect the salt flow.

2) We then simulated the neutronics of the TAP MSR with the uniform advection scheme using Moltres [10, 11]. The neutronics simulation includes the high and low power states as well as the transient ramp simulations in between these states, and is repeated for fresh fuel and fuel with equilibrium Xenon concentration.

3) The neutronics solution, which determines the heat generation profile, is then combined with the steady state fuel salt flow (velocity field) solution to obtain the thermal-hydraulics solution.

3.1 Incompressible Navier-Stokes Simulation

To simulate the Incompressible Navier-Stokes flow of the fuel salt, we used the *Navier-Stokes* physics module from MOOSE for the simulation kernels [12], with PSPG and SUPG stabilizations[1] enabled and with a stabilization factor of 1/3 for the PSPG and 1/2 for the SUPG stabilization schemes. The body force physics (gravity) was not included as the INS simulation was performed using uniform fuel salt temperature with the reactor vertically oriented and gravity would only increase axial pressure drop without significantly changing the local flow solution.

The reactor geometry used for the simulation of the INS salt flow was identical to that used in the neutronics simulation (using simple advection flow). However, while we had extended the non-flow boundary of the moderator rods into the top and bottom plenum in the simple advection case, this non-flow boundary was switched off in the INS simulation and the fuel salt could flow freely within the top and bottom plenums [See Figure 3].

For the boundary conditions, all internal surfaces were given the no-slip boundary conditions (zero velocities at the surfaces) while both the reactor inlet and outlet were given a (final) nominal axial velocity of 59.4 cm/s. This Dirichlet boundary condition value produced a slightly lower plenum axial velocity of 58.7 cm/s which corresponded approximately to the simple advection equivalent flow velocity of 65 cm/s. The discrepancy between the Dirichlet boundary condition velocity and actual plenum velocity was due to singularities at the points where the no-slip boundaries and inlet and outlet boundaries met. As mentioned previously, the simple advection equivalent flow velocity was deliberately increased from 53 cm/s in order to avoid exceeding the target outlet temperature and the temperature constraint in localized regions of the reactor core, as observed in the previous work.

Because we were simulating the core at 1/8 symmetry, the simple Dirichlet boundary condition of setting the perpendicular velocity component to be zero cannot be used on the 45° diagonal surface since the perpendicular vector at that surface was not parallel to any one of the axes. Instead, we used the *Inclined No Displacement Boundary Condition* from the *Tensor Mechanics* module in MOOSE in order to enforce

[1] PSPG: Pressure-Stabilized Petrov-Galerkin ; SUPG: Streamline-Upwind Petrov-Galerkin

the boundary condition of $\vec{u} \cdot \hat{n} = 0$. In addition to the velocity boundary conditions, the pressure at the reactor outlet was set at 0 to numerically 'pin' the pressure solution.

The key output of the INS simulation is the velocity field of the fuel salt in the core which would be used in the combined simulation discussed in section 3.3.

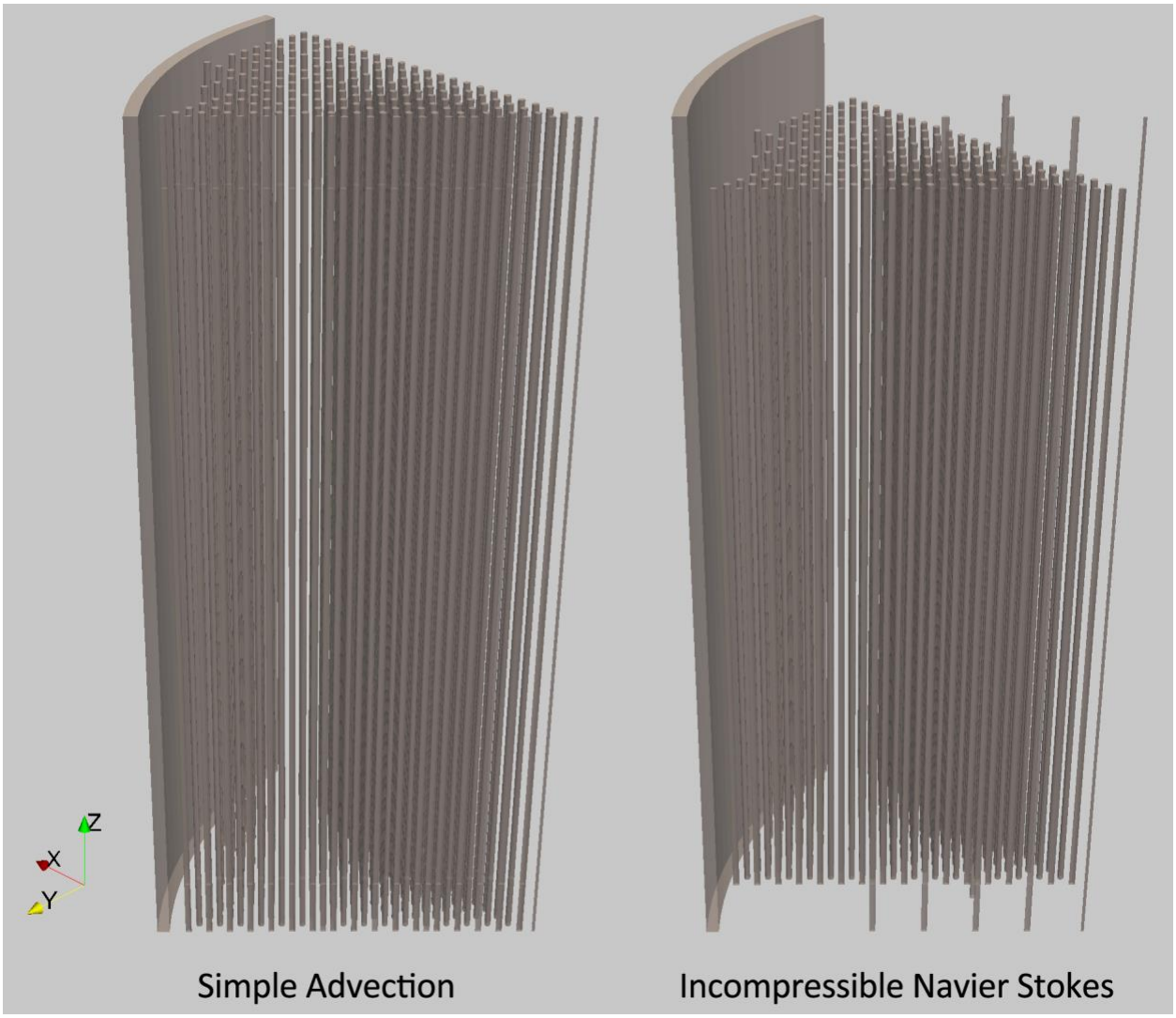

Figure 3: Geometries used for the Simple Advection simulation and the Incompressible Navier-Stokes simulation. The figure shows the non-flow reactor features with the flowing fuel salt depicted as the transparent space between the features. In the Simple Advection geometry, the plenum regions axially above and below the moderator rods were treated as non-flow regions while that in the Incompressible Navier-Stokes allow fuel salt to flow freely.

3.2 Neutronics Simulation with Simple Advection

The neutronics simulation scheme used in this work was the same as that used in the Milestone 3.1 report. However, an improvement was made in the way that the shim control was deployed.

In the previous work, the outermost control rods were inserted fully to reduce the core reactivity down to a controllable range which was managed by the central (load-following) control rod. Meanwhile, the rest of the 'unused' control rods were parked (withdrawn) in the upper plenum. In the current work however, we moved all the control rods to the same axial position to absorb the excess reactivity before using the central control rod for load-following.

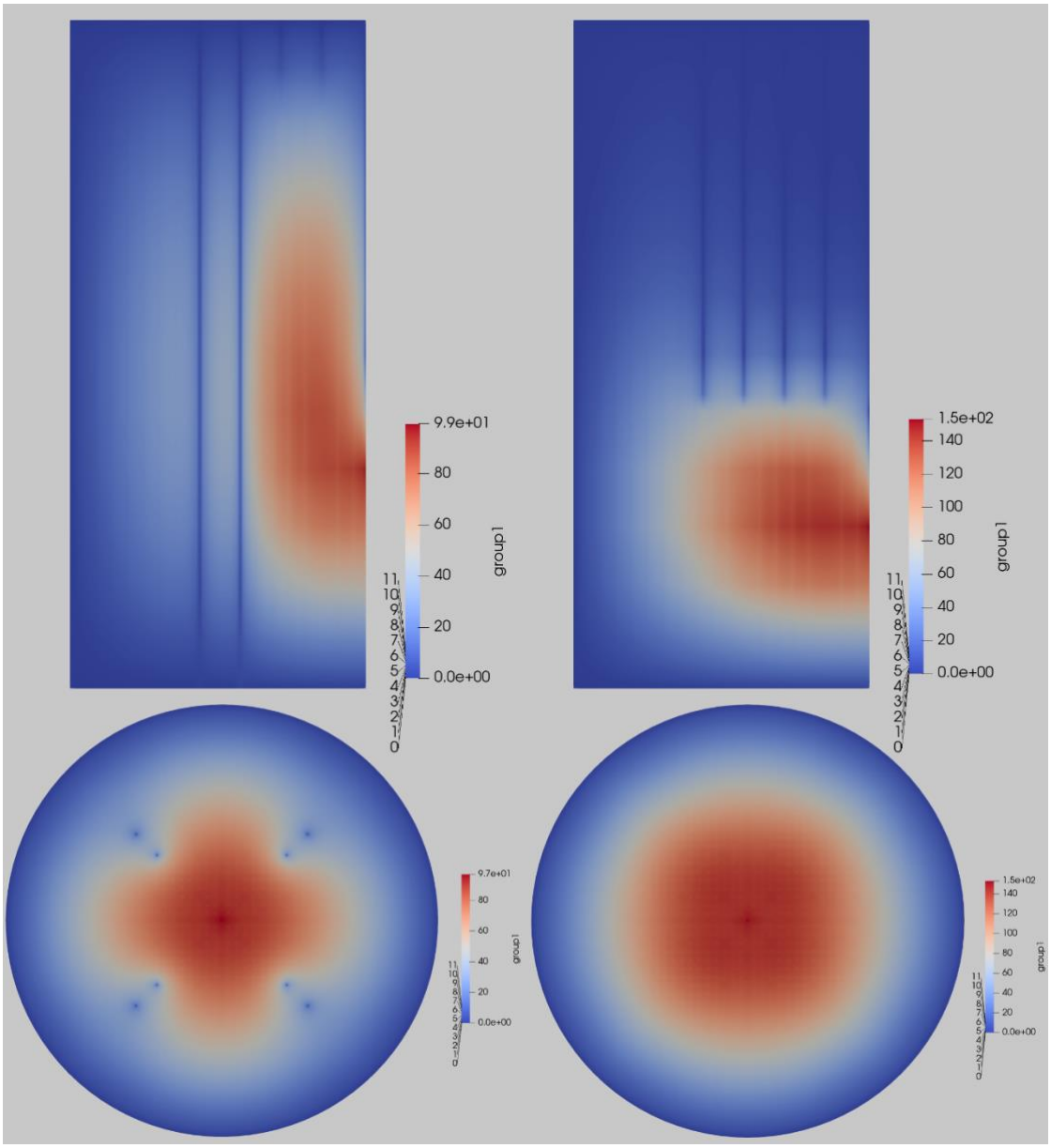

Figure 4: Neutron flux profiles (Group 1) resulting from the shim control methods used in the previous work (left) and the current work (right). The rectangular views at the top were taken from the diagonal side of the reactor in the RZ plane while the circular views were taken from top-down of the cross section of the reactor near the region of highest neutron flux. The difference in maximum neutron flux values in this figure was partially due to differences in reactor power (left: 78.8% VS right: 88.0%). The units of the scales used were in 10^{13} #/cm^3.

The difference in neutron flux profile resulting from the different shim controls is illustrated in Figure 4. In Figure 4, the axial views were taken from the diagonal side of the reactor in the RZ plane while the radial views were taken from the cross section of the reactor near the region of highest neutron flux.

In the previous shim control method, because there were two fully inserted shim rods, the neutron flux was being suppressed to form a cross-shaped region. If we require that the reactor in both configurations produce power at the same level, the heat generation in the previous configuration would be confined to the cross-shaped cross-sectional region and run the risk of overheating. Additionally, the neutron flux shape caused by the fully inserted shim rods would also cause the moderator rods behind them to become ineffective.

In the current shim control method, because all control rods other than the central rod were placed at the same axial position, the neutron flux at the active fissioning region of the core was unhindered by the control rods in the radial direction. This helped to spread out the neutron flux more evenly, leading to more uniform heat generation and as a result, less chance of overheating.

3.3 Combined Neutronics and INS Solutions for Heat Advection

The heat generation profile from the neutronics simulation was combined with the INS velocity field solution to yield the 'uncoupled' thermal-hydraulics solution. This was achieved by generating heat (temperature) using the neutron flux profiles obtained in the neutronics simulation and advecting the generated heat with the velocity field obtained in the INS simulation. The heat (temperature) advection was performed with the use of the *INSADTemperatureAdvection* kernel[2] with SUPG stabilization enabled. The SUPG stabilization was necessary due to the fact that solving for scalar advection via the Finite Element approach would often produce spurious and unphysical overshoots and undershoots if numerical stabilization was not used. Nonetheless, while greater amounts of stabilization would lead to lower artefacts and faster convergence, care was taken to use just the minimum stabilization required for the elimination of overshoots and undershoots, in order to avoid introducing numerical errors.

It should be noted that the above method would only yield the 'uncoupled' thermal-hydraulics solution where there is an absence of feedback between the neutronics and INS physics. We had chosen this approach instead of a fully coupled method due to the prohibitively high computational cost required in the fully coupled method. This topic will be elaborated further in the discussion in section 5.1.

[2] At the time of report writing, this physics kernel was made obsolete and replaced by the INSADEnergyAdvection kernel which serves a similar purpose.

4. Results and Discussion

4.1 INS Solution

The velocity and pressure solutions of the INS simulation are shown in Figure 5. In the figure, the axial velocity of the fuel salt in the central bulk region between the moderator and control rods reached approximately 90 cm/s (upwards) before decreasing towards zero near the reactor surfaces. The velocity in the bottom plenum was largely uniform except at the regions upstream of the moderator rods, where the velocity decreased towards zero at the bottom moderator rod surfaces. In the top plenum, the fuel salt formed separate high and low velocity channels downstream of the moderator rods before mixing near the top boundary due to the imposed velocity boundary condition.

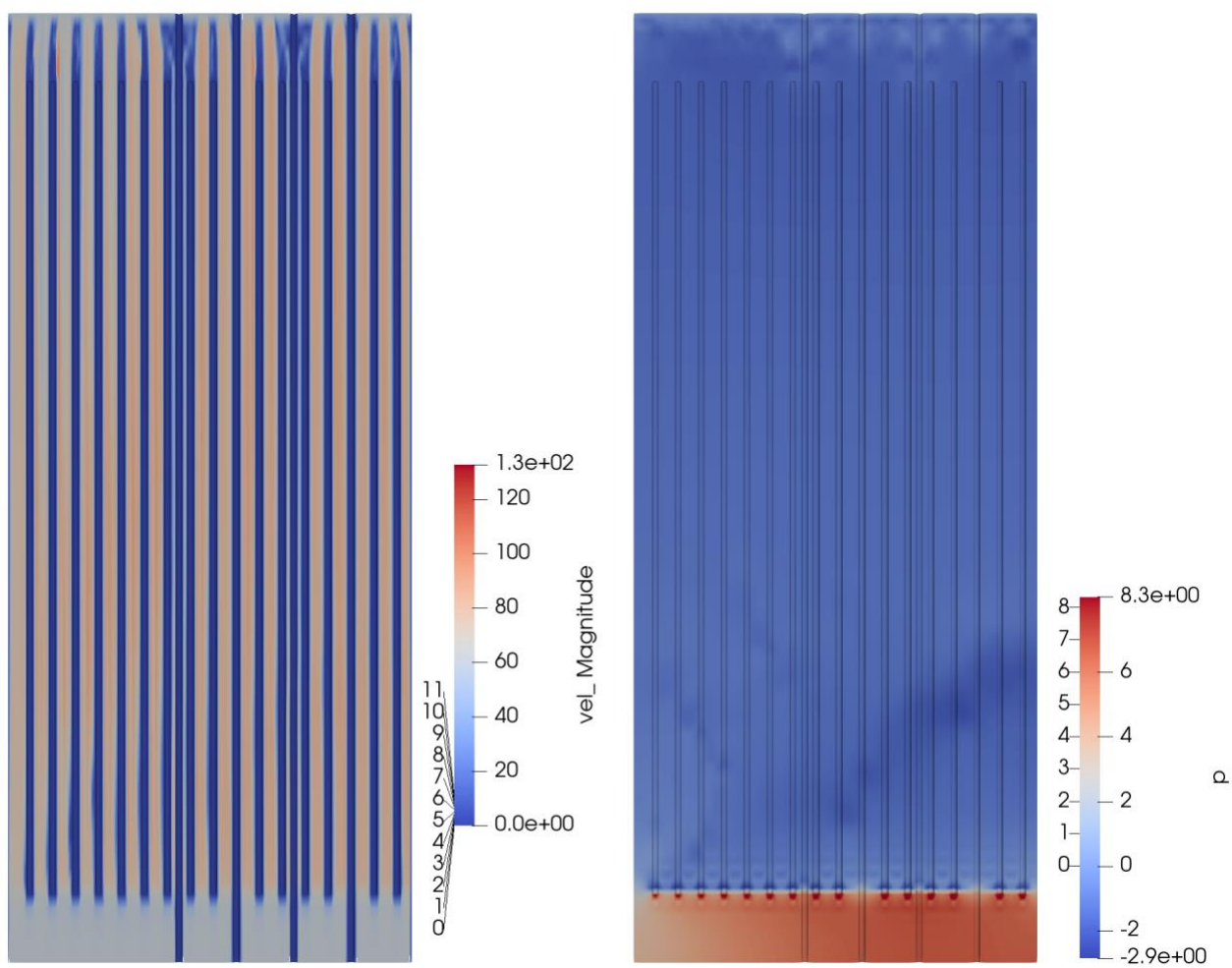

Figure 5: Velocity (left) and pressure (right) profiles from the diagonal side of the reactor in the RZ plane. In the figure, only the fuel salt was shown, with the reactor features (vessel wall, control rods, moderator rods) removed. The units of the scales used were cm/s and 10^2 Pa for the velocity and pressure profiles, respectively. The fuel salt flow in the figure was directed upwards.

In the results obtained using the INS method in this work, it should be noted that the INS physics kernels used could only solve for and obtain laminar-like solutions. As a result, the velocity field solution obtained in this method acquired laminar-like characteristics, with approximately quadratic velocity profiles between reactor surfaces which had no-slip boundary conditions. Nonetheless, the Reynolds number ($Re \approx 3300$) for the equivalent simple advection velocity suggests that the flow could have turbulent characteristics.

The discrepancy between the expected turbulent flow characteristics and laminar-like solution in the current INS velocity solution would result in two important consequences:

I. The actual velocity profiles of the fuel salt between reactor surfaces would be more uniform compared to the current INS solution due to the presence of turbulent mixing. This flattening of the velocity profiles would bring the axial velocity closer to that of the simple advection case. The turbulent mixing would also reduce the size of the low flow-velocity zone around the reactor surfaces.
II. The increased mixing from turbulence would increase the rate at which the heat near the reactor surfaces dissipates into the bulk of the fuel salt. This would reduce the maximum local temperature of the reactor core by allowing hot fuel salt to advect more readily away from the reactor surfaces.

For the pressure profile, there was a distinct high-pressure zone in the bottom plenum before transitioning to a low-pressure zone once the flow passed upwards into the reactor active region. The small amounts of negative pressure values were due to the numerical stabilization which resulted in some undershooting in the pressure solution. While the pressure profile suggests a sharp pressure drop of approximately 650 Pa as the fuel salt crosses upwards of the bottom plenum, it should be noted that the simulation was performed without gravitational forces. In practice, the hydrostatic pressure due the weight of the fuel salt ($\rho \approx 5000 kg/m^3$) would be much higher than the pressure drop observed in this simulation.

4.2 Simple Advection Neutronics Solution

The bounding neutronics solutions for the Fresh Fuel and Equilibrium Xenon cases (referred to henceforth as “Fresh” and “Xenon” for brevity) with simple advection are presented in Table 3. Table 3 summarizes the percentage power level (of the rated 1250 MW_{th}), position of the shim and central control rods, average core outlet temperature and the maximum local temperature, at the high and low power states. In these steady state solutions, the axial and radial neutron flux profiles were fundamentally similar to that shown in the Figure 4 (current shim control methods in the right diagrams).

Table 3: Summary of the reactor conditions for simulations performed with the simple advection scheme using Fresh Fuel and Equilibrium Xenon as the fuel salt. The table lists the reactor power level, shim and central control rod insertion as well as the average outlet and maximum local temperatures at the low and high power states. The up ramp and down ramp load-following operations were performed to and from these states for the respective Fresh and Xenon cases.

Case	Power Level (%)	Shim Insertion (%)	Central Rod Insertion (%)	Average Outlet Temperature (°C)	Maximum Local Temperature (°C)
Fresh (Low)	7.70	57	82	557	563
Fresh (High)	88.0	57	70	641	723
Xenon (Low)	5.97	39	70	555	561
Xenon (High)	69.5	39	62	622	704

From these high and low power steady states, the load-following simulations were performed for the Fresh and Xenon cases similar to that done in the previous work. The results of the power up ramp operations are shown in Figure 6.

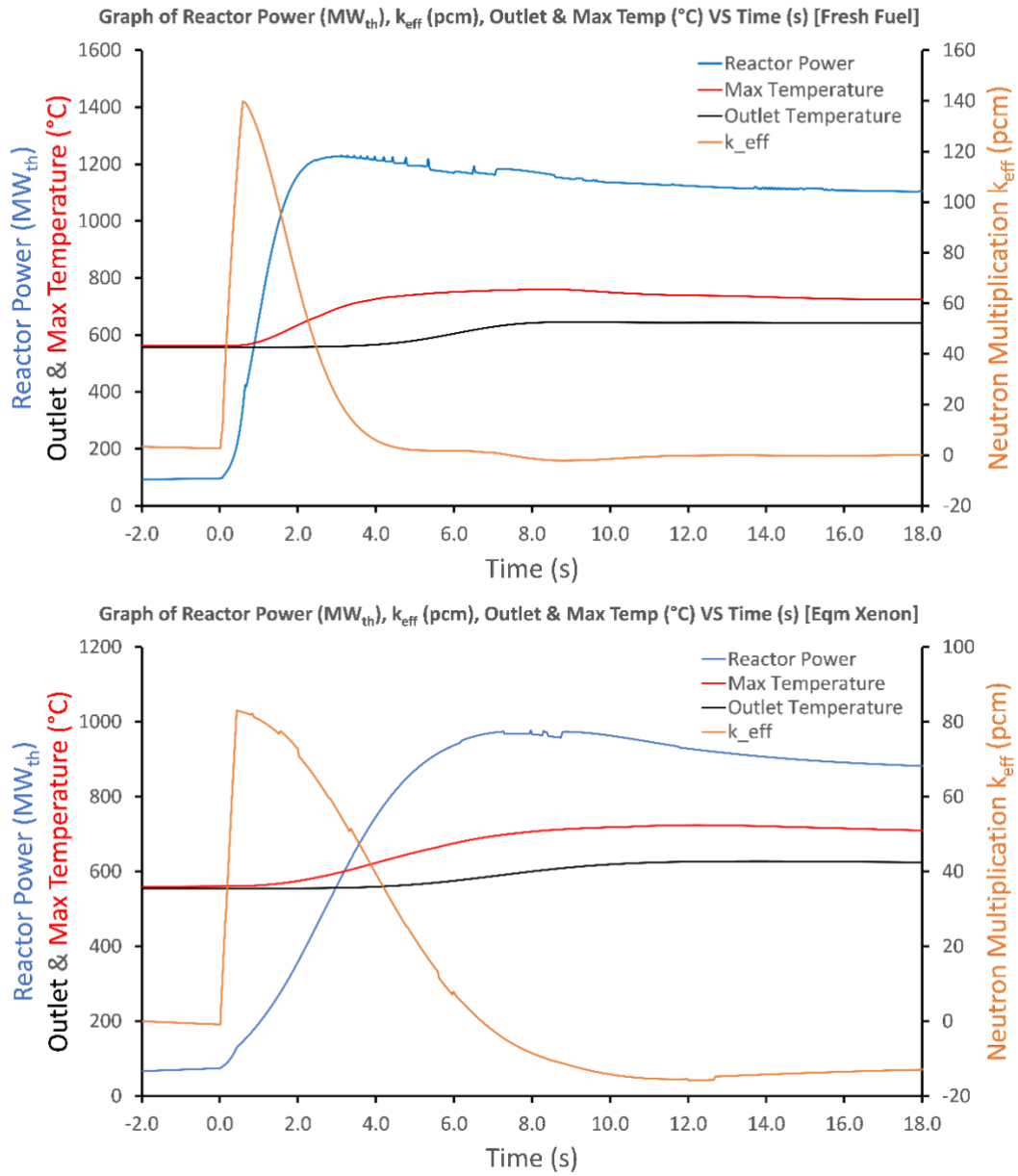

Figure 6: Graphs of Reactor Power, Neutron Multiplication Factor (k_{eff}), Outlet and Maximum Temperatures during power up ramp operations for Fresh Fuel (top) and Equilibrium Xenon (bottom). In the graph, the movement of the central control rod started at t = 0.00s and completed at 0.66s and 0.42s for the Fresh and Xenon cases respectively.

For both the Fresh and Xenon cases, the movement of the central control rod in the power up ramp operation started at time = 0s in Figure 6 and completed in around half a second. Since the graphs are approximately scaled with respect to their final power level, the graphs provide a rough indication of the transient response time for the two cases. In Figure 6, the reactor response in terms of the rise in reactor power and maximum temperature was slower in the Xenon case compared to that in the Fresh case. Because both transients occur in the order of a few seconds, the difference in response time would

produce negligible impact to power load-following operations in practice. However, the faster power response in the Fresh case could lead to a faster and larger rise in local temperatures due to a more rapid heating of the fuel salt before the increase in bulk fuel temperature could produce sufficient negative feedback. This can be observed in Figure 6 where there is a greater and faster rise in maximum local temperature in the Fresh case as compared to the Xenon case.

Since we are interested in determining whether the safety constraints of the reactor can be met during load-following operations, we will be using the more conservative Fresh case to provide the heat generation profile for the combined thermal-hydraulics solution.

4.3 Combined Thermal-hydraulics Solution

The temperature profile in the combined thermal-hydraulics solution is shown in Figure 7. In the figure, we can clearly see that the highest temperatures occurred at the walls of the reactor features and the maximum local temperature reached around 1700 K (~1400 °C). As discussed in Section 4.1, this was due to the low fuel salt velocity at the walls of these reactor features, causing heat generated at these regions to be trapped by the stagnant flow (i.e. heat in these regions mostly dissipated via thermal diffusion instead of convection). As mentioned in Section 4.1, additional mixing due to turbulence in the fuel salt flow (not included in this simulation) should help in heat removal.

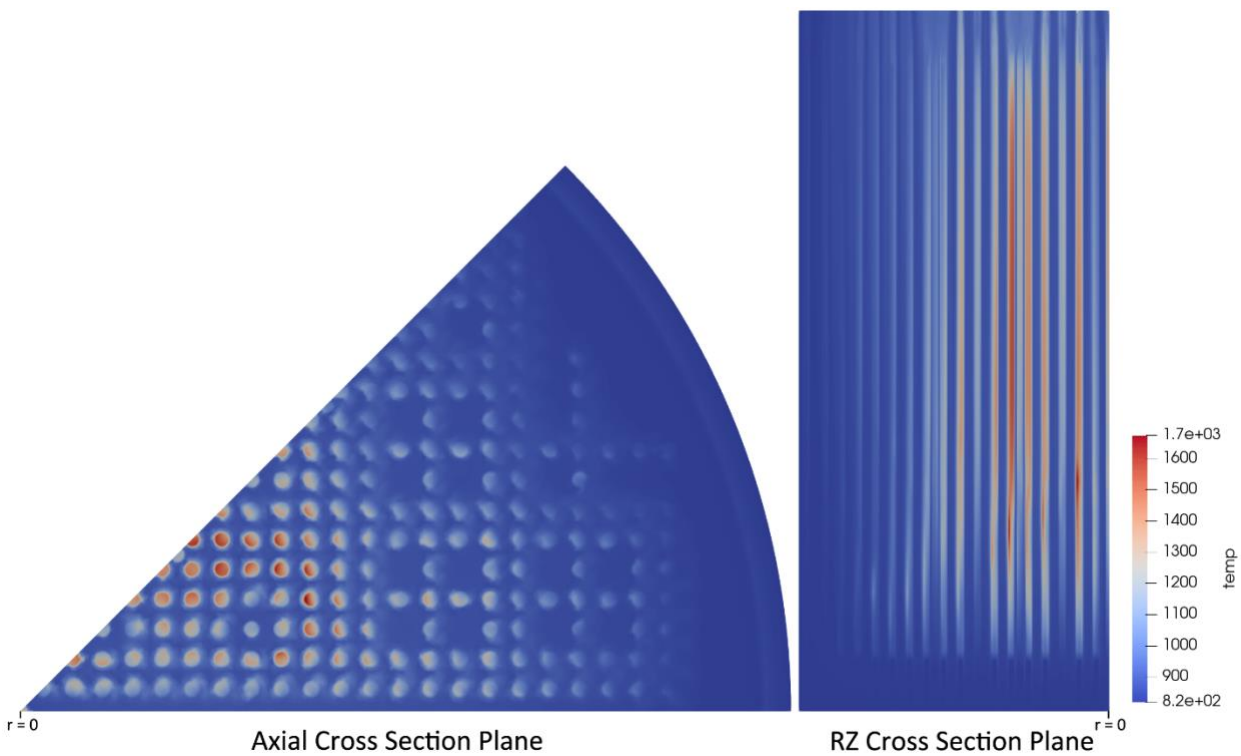

Figure 7: Axial (left) and RZ (right) cross section planes of the temperature profile solved using the solutions from the neutronics and INS simulations at t = 600s (10mins). The greatest local temperatures occur at the walls of the reactor features which correspond to the regions of low fuel salt flow in the INS solution. In the figure, the temperature scale shown was in Kelvins.

The maximum local temperature in the combined thermal-hydraulics solution is also plotted in the graphs in Figure 8. In the top graph, we can see that response time of the maximum local temperature in the combined solution is as fast as that in the simple advection case. However, while the maximum temperature in the simple advection case stabilized after around 20 seconds, the maximum local temperature in the combined solution continued increasing. From the bottom graph in Figure 8, the temperature stabilized around 200 seconds, reaching 1420 °C.

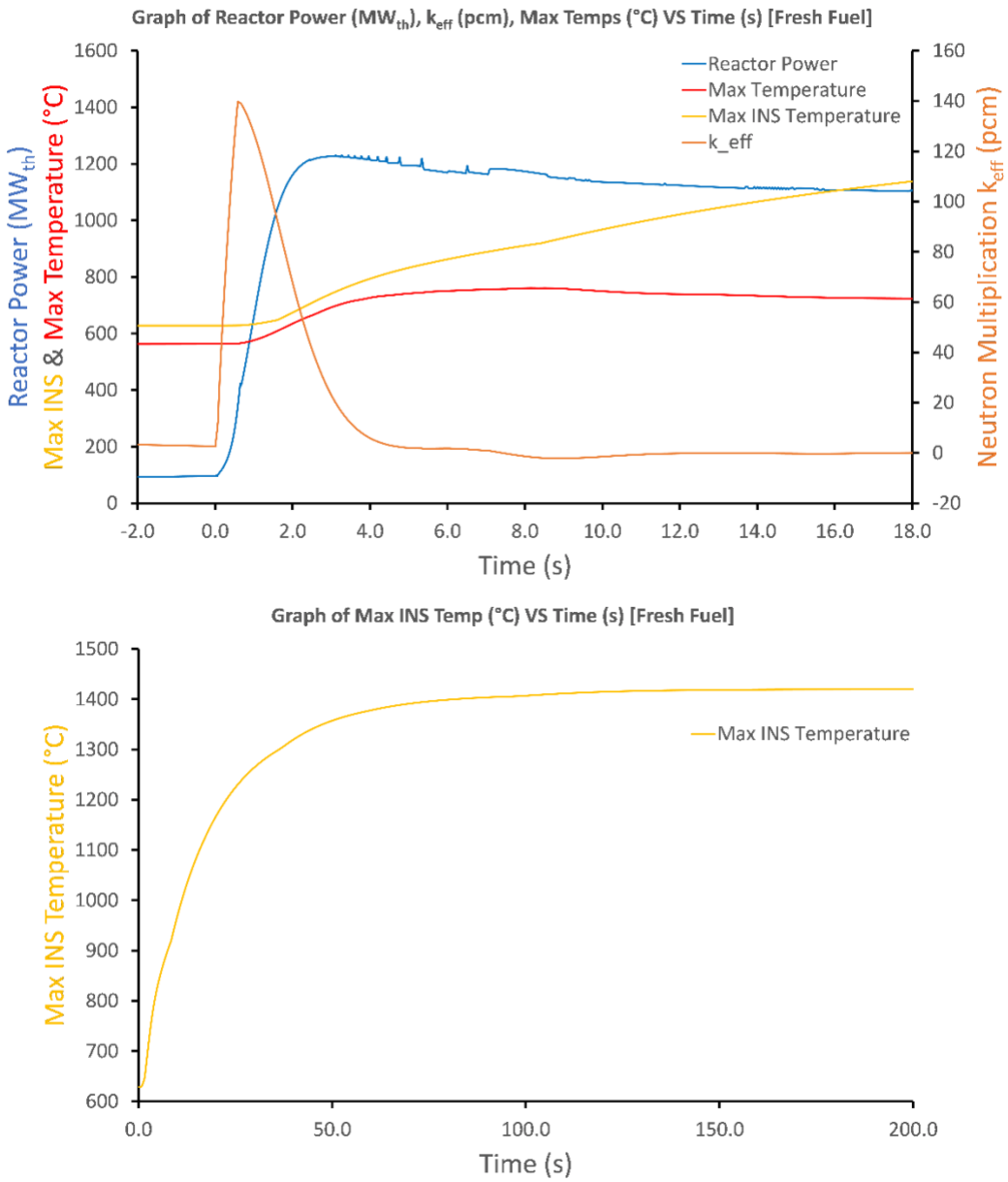

Figure 8: Graph of the maximum local temperature in the combined thermal-hydraulics solution plotted with the simple advection solution (top) for the first 18 seconds, as well as a long run plot for 200 seconds (bottom). The maximum local temperature stabilized around 1420 °C at the end of the bottom graph.

The maximum local temperature of 1420 °C obtained here clearly exceeded the temperature constraint of 900 °C for the Hastelloy components. This suggests that the simple method used here might be too conservative and warrants an alternative approach in obtaining a temperature solution for the evaluation of the safety constraint.

4.4 Unit Cell Approach

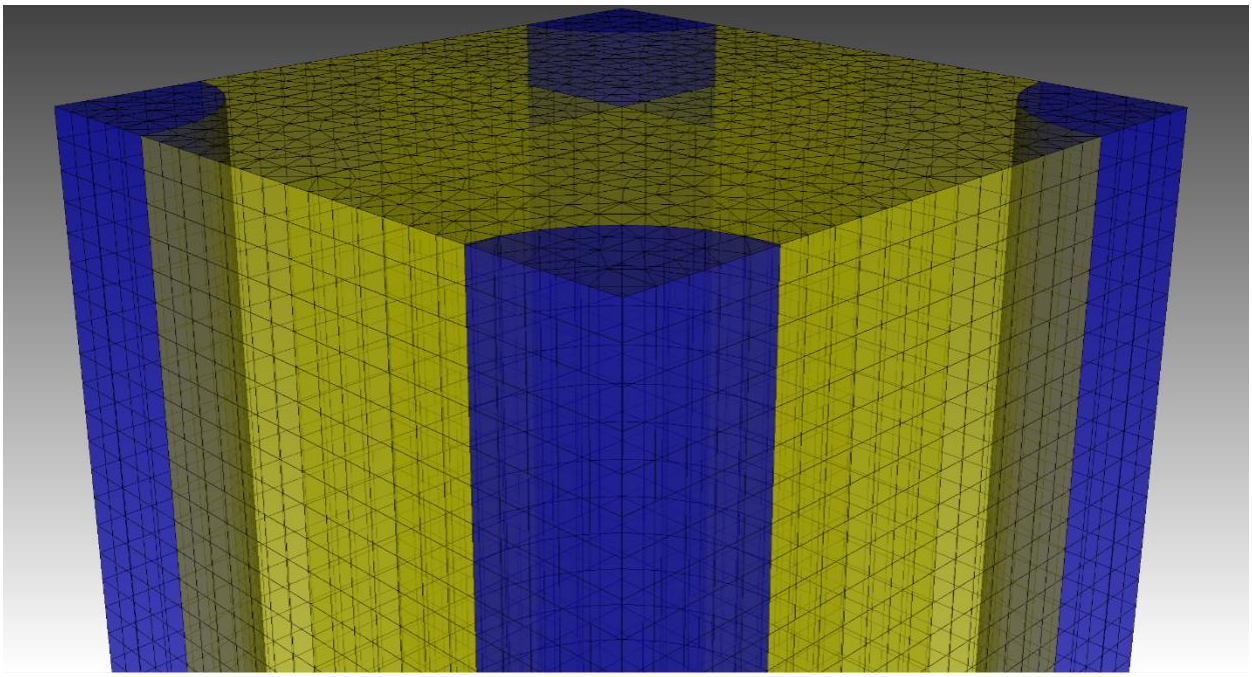

Figure 9: Side view of the full unit cell mesh. The fuel regions continuing axially from the moderator rods are shown in blue while the fuel channel is colored yellow. Due to the symmetry present in the unit cell, only one corner of the unit cell mesh and was used in the simulations.

The unit cell approach was developed in order to study the localized temperature distribution without being severely limited by the computational resource requirements of a 1/8th core simulation. The unit cell approach involved the following key steps:

1) We first generated a simplified mesh that represents a typical fuel channel in the TAP MSR as shown in Figure 9. This fuel channel consists of a central fuel region surrounded by four moderator rod quarters on each corner. Due to the symmetry present in the fuel channel, the actual mesh simulated was only one quarter of the unit cell and reflective boundaries were specified at the symmetry surfaces.

2) Axially, the unit cell consists of the top and bottom plenum regions, in addition to the central region that is partially occupied by the moderator rod quarters. To simplify the mesh further, the cladding material (SiC) was also replaced by the material of the moderator rod ($ZrH_{1.66}$).

3) We then specify the appropriate simplified boundary conditions. For heat generation, the conservative neutron flux values of $100 \times 10^{13}\ \#/cm^3$ (group 1) and $4 \times 10^{13}\ \#/cm^3$ (group 2) were used for the region between the top edge of the bottom plenum (z = -150 cm) and the bottom edge of the shim control rods (z = -21 cm).
[See Figure 10]

These values correspond to the typical values for the high power state in the Fresh case. The velocity at the inlet and outlet was set at 60 cm/s while moderator surfaces were given the no-slip boundary condition.

4) As with the earlier approach, we first solve for the steady state velocity field (~30s simulation time) before introducing heat generation physics in order to avoid overheating stagnant fuel.

The unit cell mesh size was also much finer than the 1/8th core mesh, with element size of 2.5 mm as compared to 25 mm for the 1/8th core mesh (a resolution increase by an order of 10).

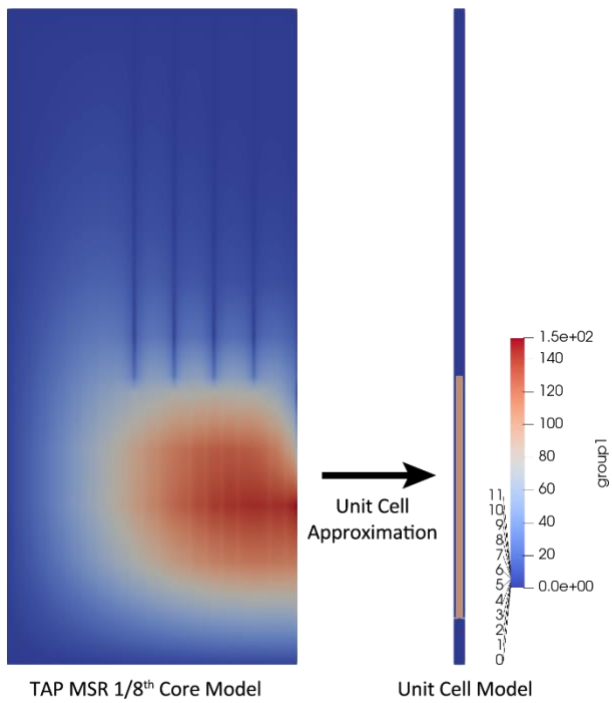

Figure 10: Diagram showing the approximation of Group 1 neutron flux in the unit cell model from the Fresh case high power state simulated using the 1/8th core model.

After obtaining the steady state velocity field solution, we turned on the heat generation physics in two slightly different conditions. In the first condition, the INS physics kernels were turned off and only temperature advection and diffusion physics were enabled to solve for the temperature evolution in the unit cell. This allows us to compare the solution obtained with the unit cell approach with the previous 1/8th core solution to determine whether mesh size had adversely affected the 1/8th core solution.

In the second condition, the INS physics kernels were left enabled and the simulation solved for both temperature advection along with the velocity and pressure fields in the presence of gravity. This is similar to the Boussinesq approach where we consider the effects of buoyancy lift caused by the lower density of the hotter fuel salt around the reactor surfaces. In this approach, when there is significant temperature difference between different regions of the fuel salt, the difference in local fuel salt densities would cause the hotter fuel salt to acquire a larger axial velocity due to buoyancy forces. This would translate to a more efficient heat dissipation leading to a decrease in maximum local temperatures.

Unit Cell Model Results:

Under the first condition (constant velocity field), the maximum temperature reached approximately 1360 °C at steady state (compared to 1420 °C in the 1/8th core model). This reduction in maximum temperature in the unit cell model suggests that the temperature gradient in the 1/8th core model was underestimated due to the larger mesh size, leading to lower heat transfer and higher temperatures in the 1/8th core model. Nonetheless, this temperature was still much higher than the desired temperature constraint of 900 °C.

In the unit cell simulation under the second condition (velocity, pressure and temperature fields were solved together), the simulation did not reach an acceptable steady state to allow for any conclusions on temperature constraints to be made[3]. Nevertheless, we observed that the velocity profile across the flow channel was continuously undulating near the moderator surfaces, as illustrated in Figure 11. This undulating of the velocity profile in turn caused the corresponding channel and maximum temperatures to fluctuate.

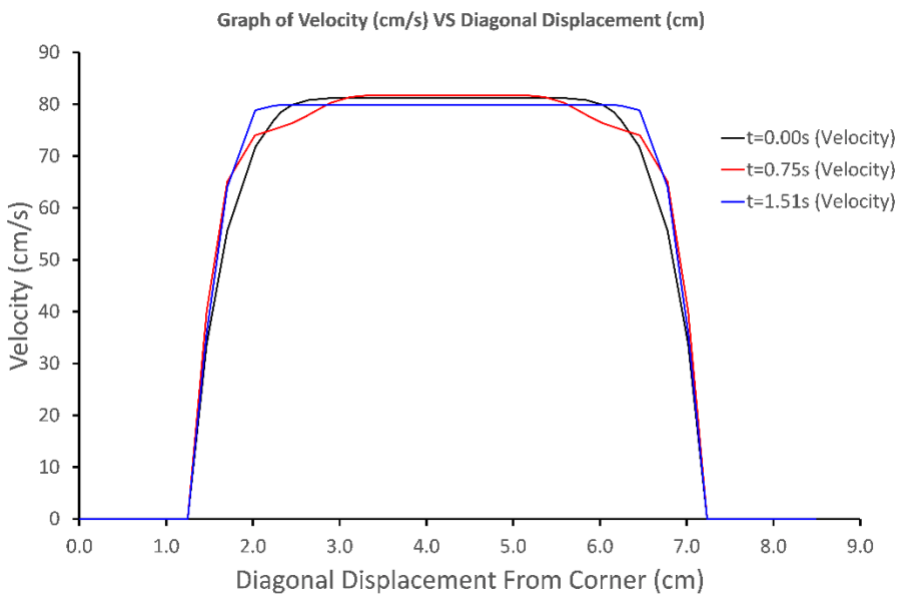

Figure 11: Velocity profiles across the fuel channel at three reference times. The figure shows the cyclical sequences in the velocity profile variation near the moderator surfaces. The changing velocity profile could be due to uneven buoyancy forces from temperature differences which disrupt the long term build up of heat near the moderator surfaces.

We postulate that the buoyancy effects due to the uneven temperature distribution provided a negative feedback where the low velocity regions (with the higher temperatures) were accelerated which caused the local temperature to decrease and in turn reduced the buoyancy before repeating the cycle. This oscillating characteristic caused the stagnating fuel to be continually swept away, disrupting the steady state near the moderator surfaces and possibly limiting the maximum temperature. In the simulation results that were obtained, the maximum fuel temperature reached a cyclical-steady value of around 720 °C, which was much lower than that observed from the previous approach. However, it should be noted that this result remains inconclusive due to the short timescale of the simulation and the fact that the simulation did not reach an acceptable steady state. With longer simulation time, we expect the velocity and temperature profiles to stabilize at a higher maximum temperature.

[3] Due to the much greater computational cost involved in the simulation.

5. Discussion and Future Work

5.1 Discussion on the Degree of Physics Coupling and Associated Difficulty

In this work, we have combined the heat generation profile from the neutronics simulation with the INS velocity field solution to yield the 'uncoupled' thermal-hydraulics solution, where heat that was generated in the fissioning region of the core moved through the core by following the velocity field. The simulation has 'no coupling' between the neutronics and INS physics because the neutronics and velocity fields were solved separately without feedback from each other.

Extending from this approach would be 'one-way coupling' where either the INS velocity field solution is used in solving for the neutronics solution or vice versa. The determination of whether to use the neutronics solution or the INS solution as the 'independent variable' depends on the relative sensitivities of the two different physics involved. In this work, we observed that the neutronics physics was much more sensitive as compared to the INS physics (i.e. the INS velocity field solution was much less sensitive to changes in power profile compared to the neutronics sensitivity to changes in the velocity profile).

Finally, if we include both the neutronics and INS physics kernels in the MOOSE/Moltres input file, we would solve both the neutronics and INS physics simultaneously and obtain a fully coupled (two-way) solution. This was the original intended approach of this study but was found to be unachievable due to the great computational cost required.

The difficulty in moving from the 'uncoupled' approach towards higher degree of physics coupling can be summarized by the following two important factors:

I. **INS mesh size**: The INS simulation requires a very fine mesh in order to accurately solve for the velocity field of the TAP MSR. This is in contrast to the mesh required by Moltres, which can tolerate relatively coarse meshes. In this work, we have refined the original mesh by a significant degree in order for the INS simulation to produce a reasonable solution. Such mesh refinement can be seen in Figure 12, where the mesh interval near the periphery had increased by a factor of 6 and the axial interval had increased from once every 30 cm to once every 2.5 cm, a factor of 12. As a result of the mesh refinement, the degrees of freedom (DOF) in the INS simulation increased from 0.33M to 4.9M (and to 2.5M in an intermediate mesh with 5 cm axial interval).

   Using 1152 cores on the INL Falcon HPC, the INS simulation took about 110 hours to reach steady state (using the 5 cm axial mesh; 127k core-hours) and an additional 6 hours for the stabilization and refinement step (using the 2.5 cm axial mesh; 7k core-hours).

II. **Difference in Number of Variables**: The INS problem contains 4 non-linear variables (pressure and the x-, y-, and z- velocity components) while the neutronics problem contains 11 non-linear variables (2 neutron flux groups, temperature and 8 neutron precursor groups). As mentioned earlier, the neutronics simulation can tolerate relatively coarse meshes, while the INS simulation requires very fine meshes. This has worked fairly well in the current work as the simulation with the larger number of non-linear variables requires a coarse mesh and vice versa, producing two separate simulations with manageable computational time.

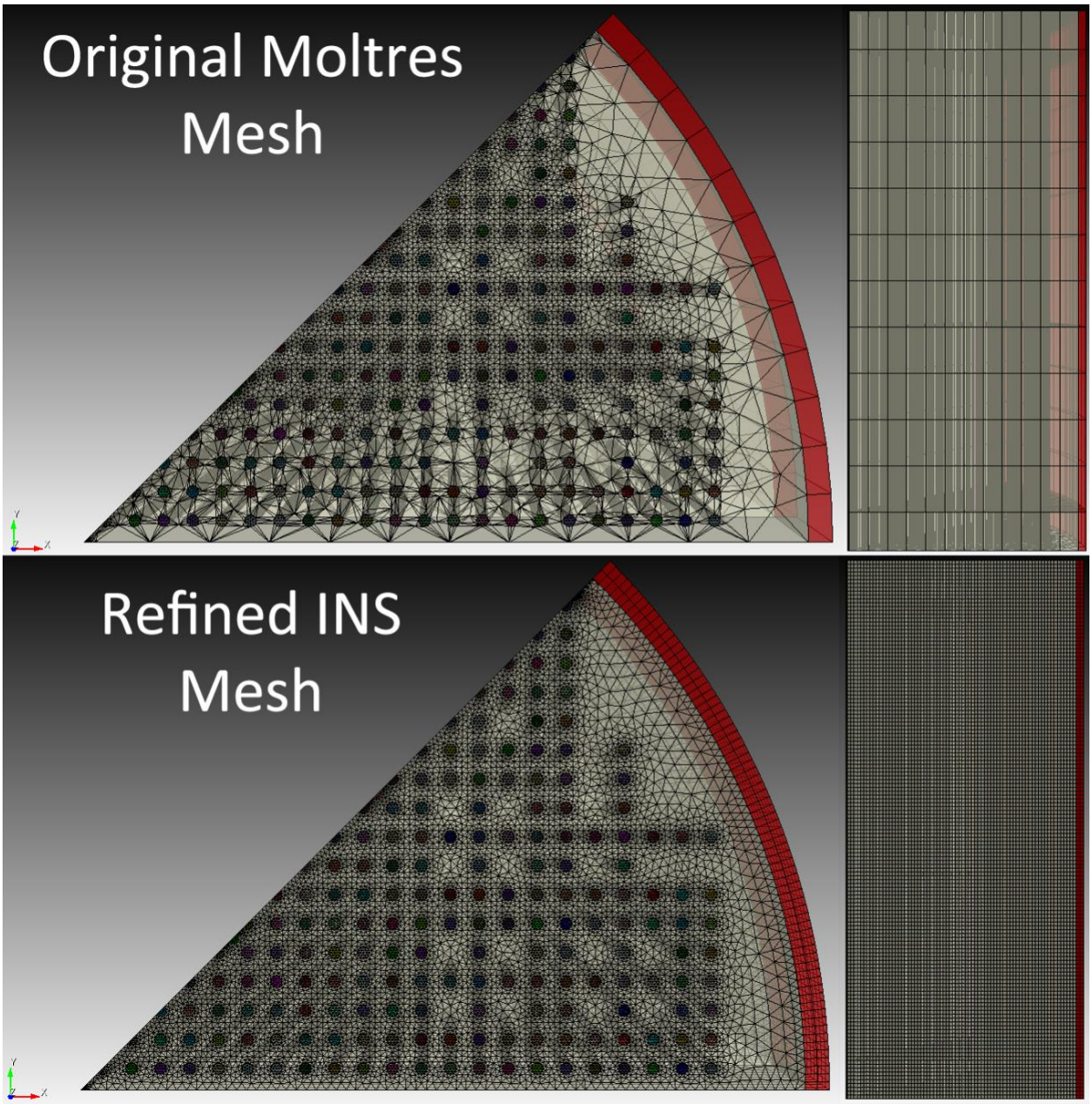

Figure 12: Axial (left) and RZ (right) cross section planes of the TAP MSR mesh, showing the difference between the original Moltres mesh and the refined INS mesh. The refined mesh shown here has a perimeter interval that is 6 times that of the original mesh, and an axial interval that is 12 times compared to that of the original mesh.

However, coupling the neutronics and INS physics into one simulation means that the neutronics kernels have to share the fine mesh required by the INS simulation. This means that the degrees of freedom would increase tremendously to an estimated 9.3M when using the 5 cm axial mesh.

Since the computational cost increases non-linearly with the increase in DOF, the simulation with the estimated 9.3M DOF is prohibitively expensive to solve.
In this work, moving from 2.5M DOF to 4.9M DOF resulted in an increase in computational time from 11 minutes to 40 minutes per timestep. If we assume that the computational cost increases in a quadratic manner with respect to DOF, then 9.3M DOF would require 160 minutes per timestep on 1152 cores, which is an unreasonably costly endeavor.

In light of the above, further physics coupling involving the 1/8th core geometry is unlikely to be practicable. In order to proceed with a higher degree of physics coupling, the size of the problem has to be reduced, such as by using homogenization or by simulating only a unit cell under typical conditions, which was done in section 4.4.

5.2 Future Work: Study of Progressively Depleted Fuel Composition

In the current and previous milestone reports, we have focused on simulations involving the beginning-of-life (BOL) fuel composition and only included a 3-day burnup fuel composition to study the effects of Xenon-135 poisoning. However, as mentioned in the previous report and in work by Rykhlevskii et al. [13], the TAP MSR at BOL operates in the epithermal neutron spectrum and is relatively unaffected by Xenon-135 poisoning. This means that the high load-following power ramp rates observed in the current and previous work only served to confirm that the TAP MSR is not sensitive to Xenon-135 poisoning at BOL.

Since we expect the TAP MSR to shift from epithermal towards thermal neutron spectrum, as more fission products accumulate which require more moderation to maintain criticality, we expect a change in the reactor's sensitivity to Xenon-135 poisoning involving progressively more depleted fuel compositions. In addition, as mentioned in the previous report, the shift in neutron spectrum is expected to impact the control worth of the reactor [14]. Therefore, it is worthwhile to determine the impact of these additional fuel composition on the load-following capability of the TAP MSR. In particular, this involves the determination of load-following ramping rates, the study of whether material safety constraints (thermal) can be met and whether sufficient shutdown margin can be maintained for these additional fuel compositions.

6. Conclusion

This work simulated the fuel salt flow of the TAP MSR using the Incompressible Navier-Stokes (INS) physics submodule in MOOSE and obtained laminar-like velocity profiles. When combined with the neutron flux profile calculated using Moltres, the thermal-hydraulics solution of the TAP MSR reached a maximum local temperature in excess of 1400 °C. An extension to the temperature study using the unit cell model with higher mesh resolution suggested that the maximum temperature might be lower at around 1360 °C in the 'uncoupled' approach and around 720 °C in the Boussinesq approach.

The thermal-hydraulics results obtained in this work remains inconclusive for evaluating the maximum fuel salt temperatures in the TAP MSR. In light of the above, we recommend that the control rod guide tubes and stabilizing structure used for securing the moderator rods be made of more temperature tolerant materials instead of Hastelloy. Our recommendation builds on top of our opinion that in the event of loss of forced convection (LOFC) accidents, the salt flow due to natural circulation would be dramatically reduced and that regions containing stagnant salt would increase, which would in turn increase the chance of surfaces reaching overly high temperatures.

Besides the thermal-hydraulics study in this work, we have also made an improvement regarding the control rods deployment, which led to greater uniformity of the neutron flux and heat generation profile. This provided an important improvement in the thermal safety of the reactor.

In future work, we will be focusing on the effects of progressively depleted fuel compositions. In particular, we will determine the load-following capability, material safety (thermal), and shutdown margin for these additional fuel compositions.

References

[1] A. J. H. Lee, T. Kozlowski and K. Huff, "Milestone 3.1 Report: Neutronics Analysis of Core Load Following Operation (UIUC-ARTS-2020-01)," Urbana, 2020.

[2] High Temp Metals, Inc, "Hastelloy C-276 Technical Data," 2015. [Online]. Available: https://www.hightempmetals.com/techdata/hitempHastC276data.php. [Accessed 1 Dec 2019].

[3] G. J. Janz, G. L. Gardner, U. Krebs and R. P. T. Tomkins, "Molten Salts: Volume 4, Part 1, Fluorides and Mixtures Electrical Conductance, Density, Viscosity, and Surface Tension Data," *Journal of Physical and Chemical Reference Data 3, 1,* 1974.

[4] E. Capelli, O. Benes and R. Konings, "Thermodynamic assessment of the LiF-ThF4-PuF3-UF4 system," *Journal of Nuclear Materials,* vol. 462, pp. 43-53, 2015.

[5] S. Yamanaka, K. Yamada, K. Kurosaki, U. Masayoshi, K. Takeda, H. Anada, T. Matsuda and S. Kobayashi, "Thermal properties of zirconium hydride," *Journal of Nuclear Materials,* vol. 294, pp. 94-98, 2001.

[6] O. Nilsson, H. Mehling, R. Horn, J. Fricke, R. Hofmann, S. Muller, R. Eckstein and D. Hofmann, "Determination of the thermal diffusivity and conductivity of monocrystalline silicon carbide (300-2300 K)," *High Temperatures High Pressures,* vol. 29, pp. 73-80, 1997.

[7] C. Wood, D. Emin and P. E. Gray, "Thermal conductivity of boron carbides," *Physical Review B,* vol. 31, no. 10, pp. 6811 - 6814, 1985.

[8] Engineering ToolBox, "Dry Air Properties," 2005. [Online]. Available: https://www.engineeringtoolbox.com/dry-air-properties-d_973.html. [Accessed 1 Dec 2019].

[9] B. R. Betzler, S. Robertson, E. E. Davidson, J. J. Powers, A. Worrall, L. Dewan and M. Massie, "Assessment of the Neutronic and Fuel Cycle Performance of the Transatomic Power Molten Salt Reactor Design (ORNL/TM-2017/475) (CRADA/NFE-16-06345)," 2017.

[10] A. Lindsay and K. Huff, "Moltres: finite element based simulation of molten salt reactors," *Journal of Open Source Software,* 2018.

[11] A. Lindsay, G. Ridley, A. Rykhlevskii and K. Huff, "Introduction to Moltres: An application for simulation of Molten Salt Reactors," *Annals of Nuclear Energy,* vol. 114, pp. 530-540, 2018.

[12] J. W. Peterson, A. D. Lindsay and F. Kong, "Overview of the Incompressible Navier-Stokes simulation capabilities in the MOOSE Framework," Idaho National Laboratory, Idaho Falls, 2018.

[13] A. Rykhlevskii, D. O'grady, T. Kozlowski and K. Huff, "The Impact of Xenon-135 on Load Following Transatomic Power Molten Salt Reactor," Department of Nuclear, Plasma, and Radiological Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, 2019.

[14] A. Rykhlevskii, J. W. Bae and K. D. Huff, "Modeling and simulation of online reprocessing in the thorium-fueled molten salt breeder reactor," *Annals of Nuclear Energy,* vol. 128, pp. 366-379, 2019.